QMEA (Problem-Solving Questions)

1. A university conducts a study on the weekly study time of its students. A random sample of 200 students reports an average study time of 14.2 hours per week, with a sample standard deviation of 6.1 hours. The population standard deviation from a previous university-wide study is known to be 6.0 hours. Of the 200 students, 116 report studying at least 15 hours per week. A previous study reported that the population variance of weekly study time was 36 hours^2.

Using the information provided above, formulate an appropriate research question for each statistical test. Then state the null and alternative hypotheses.

Statistical Test	Research Question	H_0	H_1
Z-test for a Population Mean			
t-test for a Population Mean			
Z-test for a Population Proportion			
Chi-square Test for a Population Variance			

2. A university wants to determine whether students spend less than 12 hours per week on average studying outside class. A random sample of 36 students has a mean study time of 10.8 hours and a sample standard deviation of 4.2 hours.

At the 5% significance level, test whether the average study time is less than 12 hours per week.

3. A manufacturer produces bottles with a target mean volume of 500 mL. The population standard deviation is known to be 12 mL. A random sample of 64 bottles has a mean volume of 504 mL.

At the 5% significance level, determine whether the mean filling volume differs from 500 mL.

4. A hospital records waiting times for 100 patients. The sample mean waiting time is 18.4 minutes with a sample standard deviation of 6.0 minutes. Of the 100 patients, 72 waited less than 20 minutes.

The hospital's stated service target is that patients should wait less than 20 minutes on average.

At the 5% significance level, determine whether the hospital is meeting its target.

5. A human resources manager for a car company wants to determine whether there is a difference in the number of days absent between production-line workers and office workers. Random samples of eight workers from each group were selected. The results of the statistical analysis are shown below.

Group	n	Mean	Standard Deviation
Production-line workers	8	6.25	4.27
Office workers	8	5.88	3.14

Test	Test Statistic	df	p-value
Independent t-test	0.200	14	0.844

At the 5% significance level, can we infer that there is a difference in the mean number of days absent between production-line workers and office workers?

In your answer:

1. State the null and alternative hypotheses.
2. Identify the appropriate statistical test.
3. Use the results provided to make a statistical decision.
4. State your conclusion in the context of the problem.

6. A statistics professor wants to determine whether the variability in final exam marks differs between two sections of a business statistics course taught using different teaching methods. A random sample of 11 students was selected from each section. The statistical results are shown below.

Group	n	Mean	Standard Deviation	Variance
Class 1	11	70.55	13.92	193.67
Class 2	11	75.00	7.75	60.00

Test	Test Statistic	df	p-value
F-test for Equality of Variances	3.23	(10, 10)	0.078

At the 5% significance level, can we infer that the variances of final exam marks differ between the two classes?

In your answer:

1. State the null and alternative hypotheses.
2. Identify the appropriate statistical test.
3. Use the results provided to make a statistical decision.
4. State your conclusion in the context of the problem.

7. Surveys are widely used by politicians to monitor public opinion. Six months ago, a survey of 1,100 voters found that 56% supported a national party leader. This month, another survey of 800 voters found that 46% supported the leader. The statistical results are shown below.

Survey	Sample Size	Sample Proportion
Six months ago	1,100	0.56
This month	800	0.46

Test	Test Statistic	p-value
Two-Proportion Z-test	4.30	< 0.001

At the 5% significance level, can we infer that the national leader's popularity has decreased?

In your answer:

1. State the null and alternative hypotheses.
2. Identify the appropriate statistical test.
3. Use the results provided to make a statistical decision.
4. State your conclusion in the context of the problem.